

Answers:

1. you need to mass out 5.0127 g of FeSO_4 into 150 mL of water in a beaker and stir to dissolve 2.a) 2.625 mL b) 9.33 mL
3. 17.41 mol/L 4. 0.145 mol/L 5. 84.0 mL
- 6.a) both: made in a volumetric flask, used to find the concentration of another solution, react with each other according to stoichiometry, both form stable solutions
b) if one is an acid, the other is a base; primary standard must be as pure as possible but titrant does not have to be as pure; primary standard must be as solid at SATP, but titrant does not have to be a solid at SATP; primary standard must not be hygroscopic or gas off, but titrant can be hygroscopic or can gas off; primary standard should form a colourless solution, but titrant does not have to form a colourless solution; primary standard must have small crystals, but titrant does not have to be small crystals
- 7.a) the moles of H^+ from the acid = the moles of OH^- from the base ($\#_{\text{OH}^-} \cdot M_B \cdot V_B = \#_{\text{H}^+} \cdot M_A \cdot V_A$)
b) first sign of permanent colour change in the indicator (approximately = end-point, but are not the exactly same thing)

Solutions:

1. $n = MV = (0.22)(0.15) = 0.033$ moles; $m = n(\text{mm}) = 0.033(151.9) = 5.0127$ g
- 2.a) $M_1V_1 = M_2V_2$ so $12V_1 = (0.14)(225)$ and $V_1 = (0.14)(225)/12 = 2.625$ mL
b) $M_1V_1 = M_2V_2$ but $M_2 = (3/4)M_1 = (0.75)(4.6) = 3.45$ = a 25% dilution in concentration; so now $(4.6)(28) = 3.45V_2$;
 $V_2 = (4.6)(28)/3.45 = 37.33$ mL of solution; $V_2 = V_{\text{water}} + V_1$; so $V_{\text{water}} = V_2 - V_1 = 37.33 - 28 = 9.33$ mL of water added
3. S.G. = m/V is really density (know by units) so assume 1000 mL of solution (density is a ratio); so S.G. = m/V ; so
 $m = \text{S.G.}(V) = 1.0496(1000) = 1049.6$ g; now assay = $m_{\text{solute}}/m_{\text{solution}} \times 100\%$; so $m_{\text{solute}} = (\text{assay})(m_{\text{solution}}) = 0.995(1049.6) = 1044.352$ g; then $n = m/\text{mm} = 1044.352/60.0 = 17.41$ mole and we assumed 1000 mL = 1 L which means we have concentration as $M = n/V = 17.41/1 = 17.41$ mol/L
4. $\text{Bi}_2\text{S}_3(\text{s}) + 6\text{HCl}(\text{aq}) \rightarrow 2\text{BiCl}_3(\text{s}) + 3\text{H}_2\text{S}(\text{aq})$; $n = m/\text{mm} = 3.8/315.5 = 0.012$; 2 mol $\text{BiCl}_3(\text{s})$ produced with 3 mol $\text{H}_2\text{S}(\text{aq})$
 $V = 125$ mL $m = 3.8$ g $M = ?$ 0.012 moles produced with x
 $x = 3(0.012)/2 = 0.0181$ mol; $M = n/v = 0.0181/0.125 = 0.145$ mol/L (all the same volume)
5. $\#_{\text{OH}^-} \cdot M_B \cdot V_B = \#_{\text{H}^+} \cdot M_A \cdot V_A$; $(2)(1.8)V_B = 3(1.4)(72)$; so $V_B = 3(1.4)(72)/(2)(1.8) = 84.0$ mL